

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 38

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Spearman's rank correlation coefficient (1) • Chi-squared test (frequencies / counts in categories) (1) • t-test (two means of measured data) (1) • Spearman's rank correlation coefficient (1)	4
01.2	<i>1 mark.</i> • Chi-squared compares frequencies / counts in categories; it cannot be used for continuously variable measured data (1)	1
01.3	<i>1 mark for each point.</i> • The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level (1) • The negative sign shows a negative correlation: one variable decreases as the other increases (1) • There is less than a 5% probability the correlation arose by chance; reject the null hypothesis (but correlation is not proof of causation) (1)	3
01.4	<i>1 mark for each point.</i> • Show the standard deviation / range of the original values around each mean (1) • If the bars of two means overlap greatly the difference is probably not significant; little overlap suggests a significant difference (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) (1) • LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia (1) • Herbivores are more energy-efficient (lower trophic level) so more food per unit area (1)	3
02.2	<i>1 mark for each point.</i> • Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm (1) • Crossed with normal females (XX) all offspring are XX females (1) • Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently (1)	3
02.3	<i>1 mark for each point.</i> • Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation (1) • Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density (1) • Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification (1) • Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
03.1	<i>1 mark for each point.</i> • Other MEDCs (USA, Canada, Australia) also had surpluses; Japan already had suppliers (1) • Eastern Europe / USSR needed food but could not pay (1) • Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production (1)	3
03.2	<i>1 mark for each point.</i> • Quotas limiting production (eg milk) (1) • Set-aside payments for taking land out of production (1) • Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) (1) • Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming (1)	4
03.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape (1) • Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat (1) • Beetle banks of tussocky grass - natural pest predators (1) • Skylark nesting plots; wild bird seed / nectar plants - winter food and pollinators (1) • In-field tree protection; controlled stocking density to reduce compaction; ditch management; dry-stone walls; woodland management (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession (1) • Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes (1)	2
04.2	<i>1 mark for each point.</i> • Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out (1) • Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles (1) • Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes (1) • Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species (1)	4
04.3	<i>1 mark for each point.</i> • Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass (1) • Choice depends on the vegetation to be controlled and the plants to be protected (1)	2
04.4	<i>1 mark for each point.</i> • Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground (1) • Heat stimulates dormant heather seeds to germinate so young heather re-establishes (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	
05.1	<i>1 mark for each point.</i> • Harvest rate below the maximum sustainable yield; only economically valuable species (mahogany) selectively logged, not clear-felled (1) • Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing; large seed trees and small trees left to grow (1) • Buffer zones with no harvesting or machinery near rivers / wetlands (1) • Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees (1) • Each felled tree tagged and tracked to the retailer for FSC accreditation; rangers control illegal logging; tree nurseries replant over-exploited and ecologically important species (1)	5
05.2	<i>1 mark for each point.</i> • The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it (1) • Employment as rangers, forestry and tourism workers; ecotourists use local restaurants, produce and souvenirs; offcuts given to local carvers (1) • Benefits give the community an incentive to protect the forest; carbon sequestration payments from companies fund management (1)	3

Question	Marking guidance	Marks
05.3	<i>1 mark for each point.</i> • Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept; timber resource used over a longer period (1) • Disadvantage: more labour-intensive and expensive; more tracks needed per tree (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	
06.1	<i>1 mark for each point.</i> • Warming melts ice shelves that hold back glaciers and lubricates ice flow so land ice reaches the sea faster (Antarctic Peninsula most affected); sea ice loss removes krill habitat and lowers albedo (1) • Raised UV harms planktonic algae, krill and fish / sea urchin larvae in surface waters (land animals protected by fur / feathers) (1) • Disturbance of wildlife, introduction of non-indigenous species and pathogens, fuel and sewage pollution; visitor numbers rising rapidly at a few landing sites (1) • Reserves of coal, oil, gold, uranium, platinum etc; pressure to exploit will rise as reserves elsewhere deplete though currently banned and operations extremely difficult (1)	4
06.2	<i>1 mark for each point.</i> • Treaty signed by 52 countries including all active there; no military activity (except supporting research), no nuclear explosions or radioactive waste dumping (1) • No commercial mineral exploitation (1) • Fisheries regulated by CCAMLR: krill and some fish exploited, fisheries closed if over-exploited; no commercial exploitation of mammals (1) • Tourism control: no accommodation on land (wastes stay on ships), one ship per site with maximum 100 ashore, trained guides, keep 5 m from wildlife, no litter, cleaned protective clothing to prevent introductions (1) • Waste management: all wastes removed (fuel containers, batteries, plastics, oil); sewage only discharged where sea currents disperse it; old waste removed unless removal causes more damage (1) • Only long-term residents are research scientists and support staff; strict regulation of research impacts (1)	6
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 69-71; AQA spec 3.1.2.3.4.	
07.1	<i>1 mark for each point.</i> • Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds (1) • Water-repelling surface of lotus flowers (1) • Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to (1) • Hooked burrs on seeds that stick to animal fur (1)	4
07.2	<i>1 mark for each point.</i> • The mound absorbs sunlight and warms; hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) (1) • Air blowing over the mound has a lower pressure which also draws air out (1) • Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used (1)	3

Question	Marking guidance	Marks
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marsupials: young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems (1) • Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns (1) • Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs (1)	2
07.4	<i>1 mark.</i> • They are light but strong: hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	
08.1	<i>1 mark.</i> • Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) (1)	1
08.2	<i>1 mark for each point.</i> • Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant (1) • Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield (1)	2
08.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Replacement of valuable animals that die with genetically identical individuals (1) • Replacement of herds culled during a disease outbreak (1) • Production of large numbers of individuals with desirable characteristics (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
09.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands; HEP schemes alter river levels flooding sandbank nests of turtles / birds; species with narrow tolerance (specialists) affected first; sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade (2) • Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae (2) • Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish) (2)	6
09.2	<i>1 mark for each point.</i> • Reduces light penetration so submerged plants cannot photosynthesise (1) • Blocks the gills of filter feeders such as bivalve molluscs; sediment smothers the bed (1)	2
09.3	<i>1 mark for each point.</i> • Rare species often have very specialised habitat requirements / narrow ranges of tolerance (1) • A small change moves conditions outside their range while generalists survive; small populations cannot recover (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes • Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time • Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management • Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
	Model answer - Level 3 (8 marks)	
	Succession is the sequence by which pioneers colonise bare ground, modify the abiotic conditions and are replaced by later seres until a climax community controlled by climate is reached; where grazing, mowing, burning or coppicing halts it, a plagioclimax forms. This tells a conservation manager whether to intervene: chalk grassland, heathland and coppiced woodland are plagioclimaxes that disappear without continued management, whereas a climax habitat is best left alone, and habitat creation must plan the stages, control scrub and manage secondary succession.	

Question	Marking guidance	Marks
	Population dynamics - birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles and r- and k-selection - inform decisions of a different kind. Maximum sustainable yield sets harvest levels in fisheries, whaling and forestry; k-selected species that breed late and slowly need strict protection and recover slowly; culling may be justified where predators are absent, as with Scottish red deer without wolves, or where introduced species threaten natives, while the reintroduction of wolves to Yellowstone shows how restoring a predator changes vegetation through the food web. Monitoring population size, distribution, survival and age structure measures success, and knowledge of adaptation and ranges of tolerance guides how abiotic conditions are managed.	
	The limits are uncertainty: natural fluctuations, unknown carrying capacities and the difficulty of predicting how communities respond. Ecological understanding cannot remove that uncertainty, but without it conservation decisions - to manage or not, to cull or protect, how much to harvest - are guesses.	
	Weaker answer - Level 2 (5 marks)	
	Succession is when a habitat changes over time from pioneer species to a climax community. Conservationists need to understand this because some habitats like heathland will turn into woodland if they are not managed by grazing or burning. Population dynamics is about how populations grow and are limited by factors like food and predators. This helps conservation because you can work out how many animals can be taken without the population declining, and when a population needs protecting. Monitoring populations shows if conservation is working.	
	What makes the difference	
	• Level 2 explains both concepts correctly and links them to decisions, but generally; Level 3 uses the technical vocabulary (seres, plagioclimax, k-selection, MSY) with named cases (Yellowstone wolves, Scottish deer, chalk grassland).	
	• Level 3 covers more applications (habitat creation, culling introduced species, adaptation and tolerance) and evaluates the uncertainties.	
	• Structure: Level 3 gives each concept its own paragraph with the decisions it informs, then a conclusion.	
10.2	<i>1 mark.</i> • Any taxonomic group of organisms classified together by similarity (eg species, genus, family) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Question	Marking guidance	Marks
11.1	Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Sustainability definitions; population and per capita impact; attitudes of rural and urban societies • Negative feedback: cloud cover and albedo, CO₂ and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • Tipping points: rare under negative feedback; large events (Permian methane hydrate); self-sustaining change from human activities • Diversity and resistance: reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides); links to climate change chapter; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	25

Question	Marking guidance	Marks
11.2	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradications, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25